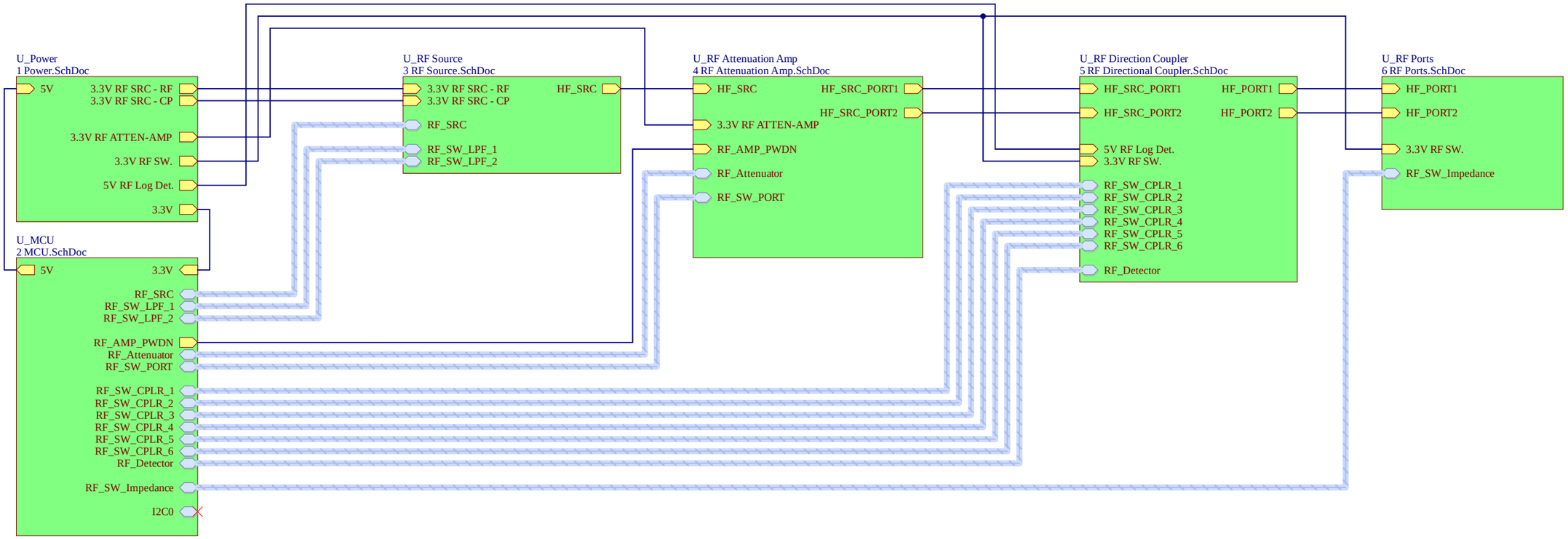
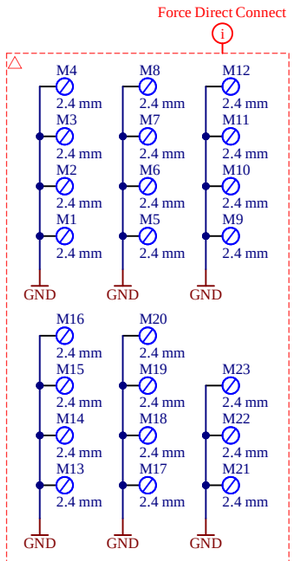


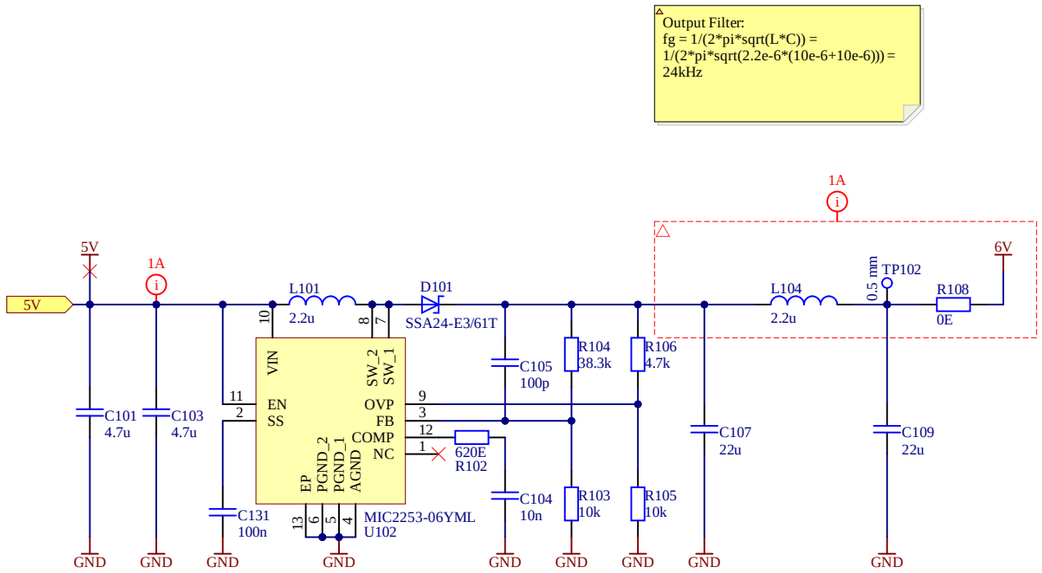


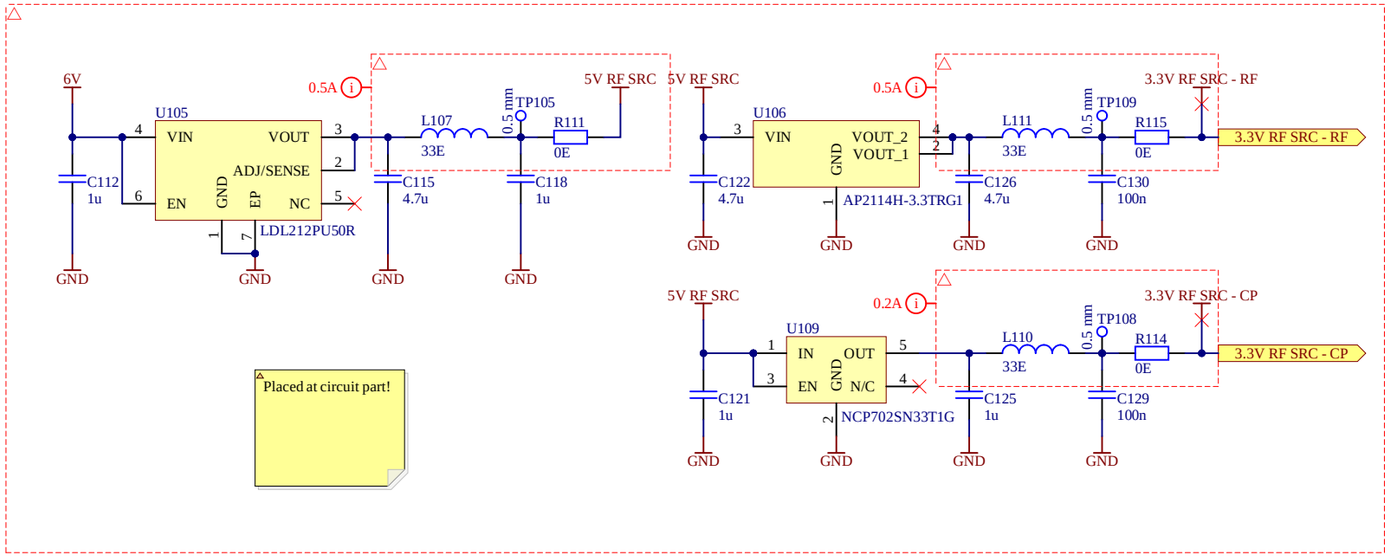
VERTRAULICH CONFIDENTIAL Copyright by Spacetek Technology AG. Für dieses Dokument behalten wir uns alle Rechte vor. We reserve all rights in respect of this document.

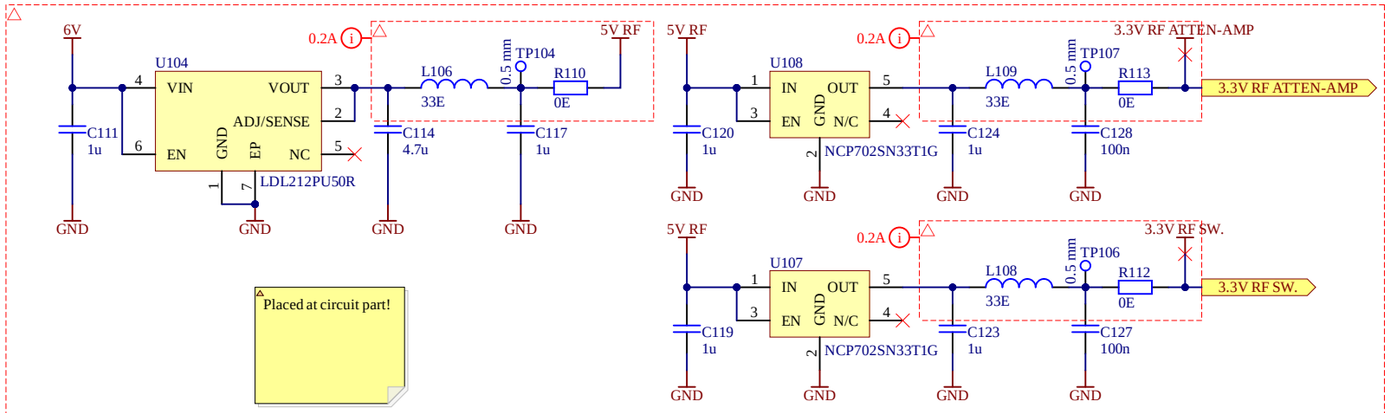
Spacetek Technology AG | Gibb
Spacetek, Gümligen
Gibb, Bern

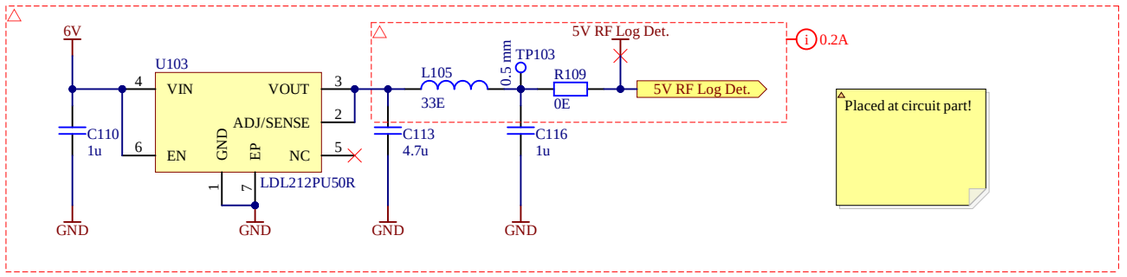
Project:	SNA	**Number:**	SNA	**Git:**	1524f681
Board:	SNA	**Variant:**	EMS	**Version:**	1.0
Page:	Overview	**Release:**	03.09.2026	**Sheet** 1 of 7	**Design:** ML
				Size: A3	**Check:** RM

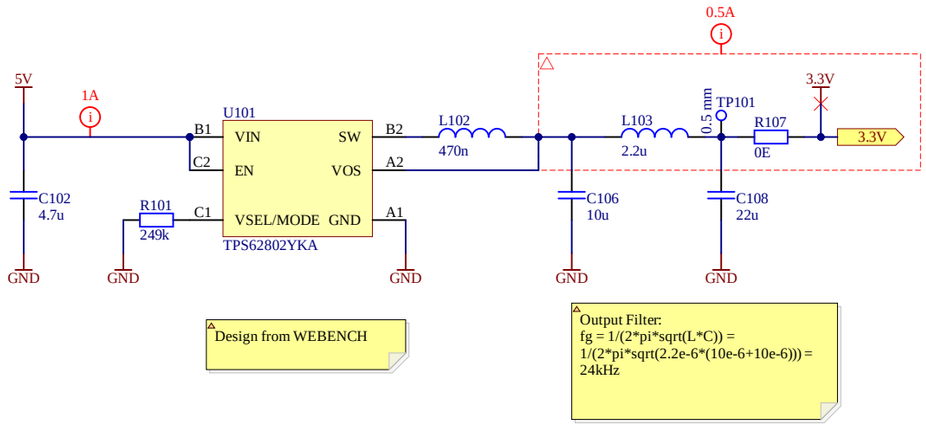
File: https://github.com/ml3rc/SNA.git 0 Overview.SchDoc











L = 2.2uH as per reference design
Cout = 22u as per reference design

Set ss to 10ms
Css=(Tss - 1ms) / 85kOhm = (10ms - 1ms) / 85kOhm = 105nF -> 100nF (10nF internal cap!)

Uovp = 8V
Rbottom = 10k
Rtop = (Uovp*15kOhm*Rbottom)/(1.245*67kOhm)-Rbottom = (8V*15kOhm*10kOhm)/(1.245*67kOhm)-10kOhm = 4.39kOhm -> 4.7kOhm

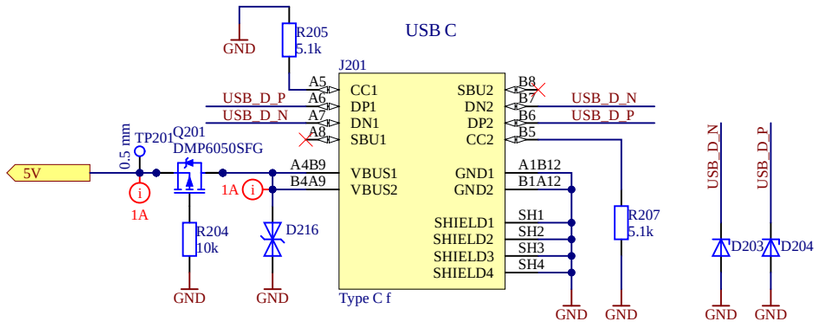
Uout = 6V
Ufb = 1.245V
Ifbin = -450nA
Rbottom = 10kOhm
Rtop = (Uout - Ufb)/(Ufb/Rbottom+Ifbin) = (6V-1.245V)/(1.245V/10kOhm-450nA) = 38.3kOhm

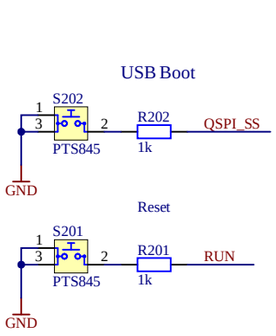
Ccomp = 10nF as per reference design
Rcomp = 620Ohm as per reference design

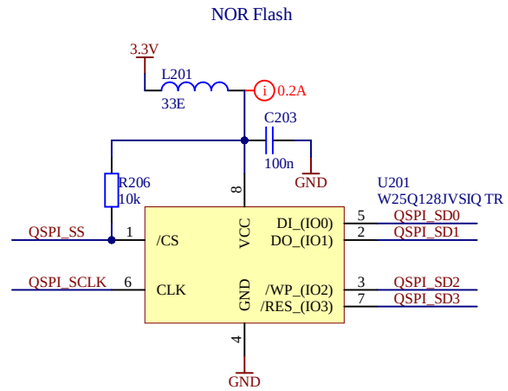
For current specs see power blockdiagramm
20mil -> 1A (2.14A)
15mil -> 0.5A (1.79A)
12mil -> 0.2A (1.56A)

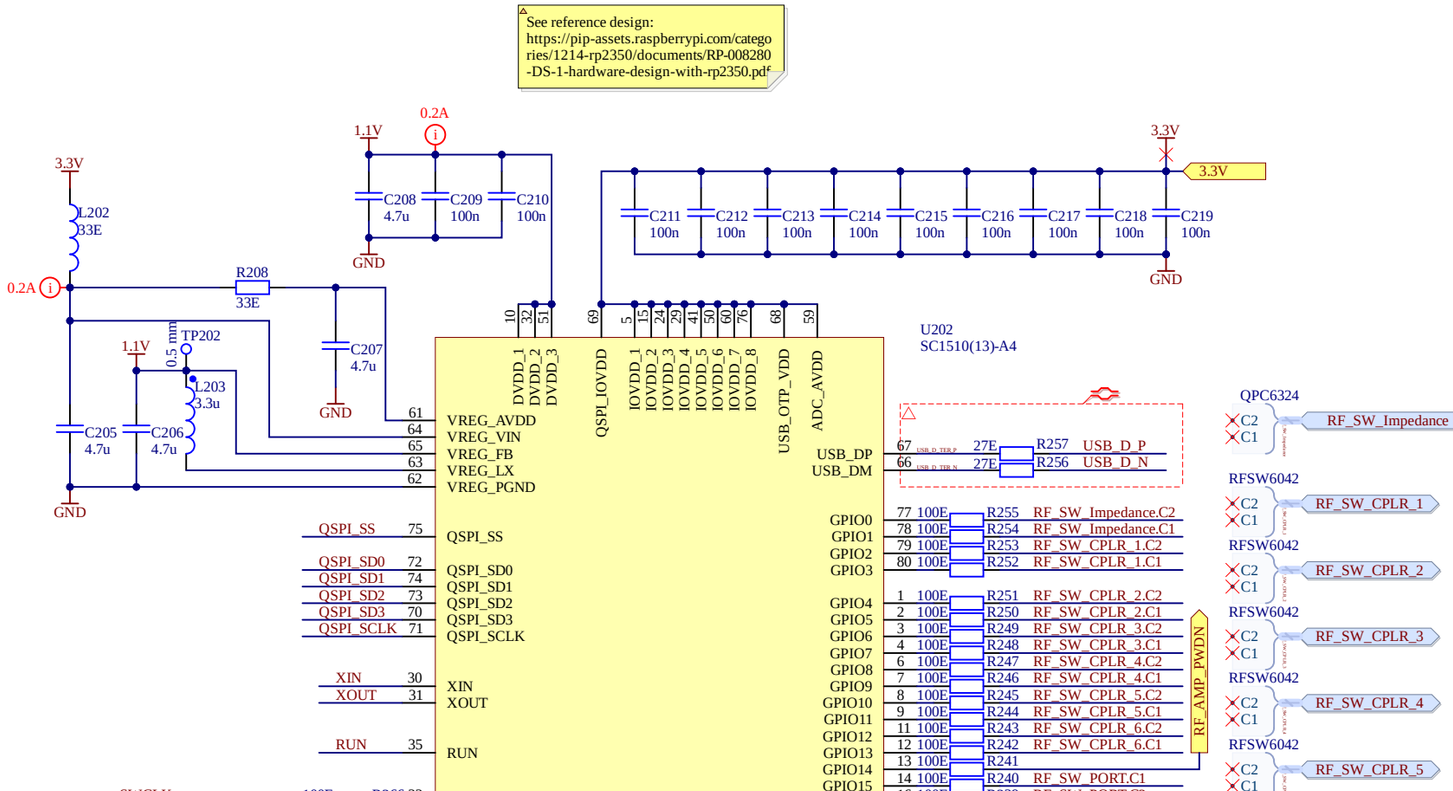
VERTRAULICH CONFIDENTIAL Copyright by Spacetek Technology AG. Für dieses Dokument behalten wir uns alle Rechte vor. We reserve all rights in respect of this document.

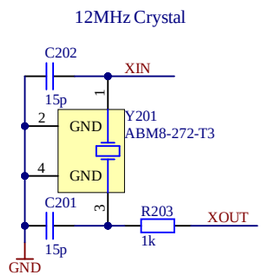
Spacetek Technology AG | Gibb Spacetek, Gümligen Gibb, Bern	**Project:** SNA **Board:** SNA **Page:** Power	**Number:** SNA **Variant:** EMS **Release:** 03.09.2026	**Git:** 77bd9c41 **Version:** 1.0 **Sheet** 2 of 7 **Design:** ML **Size:** A3 **Check:** RM
		File: https://github.com/ml3rc/SNA.git	1 Power.SchDoc

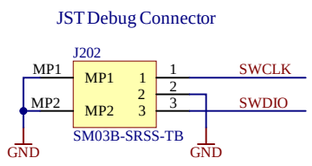


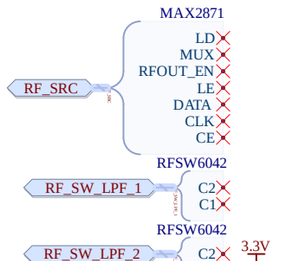


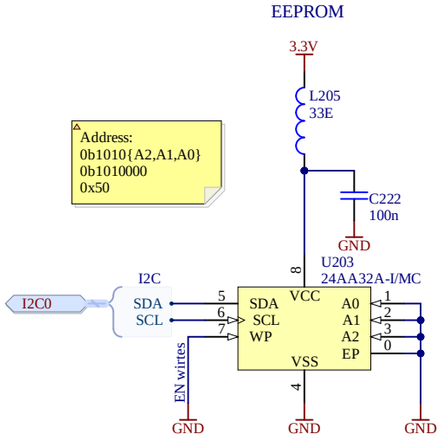


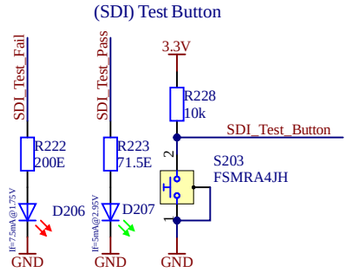


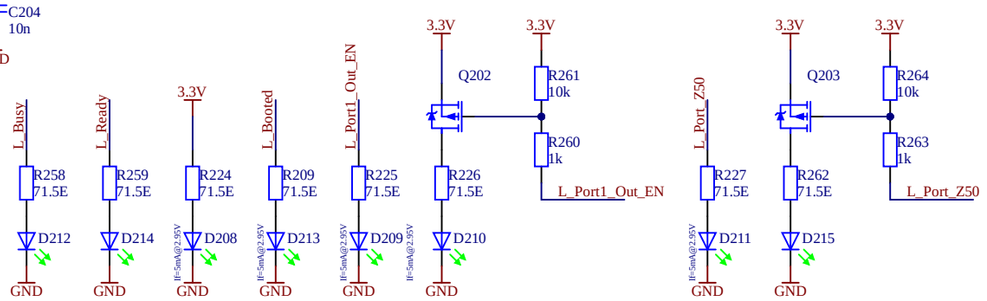


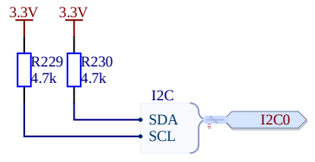












VERTRAULICH CONFIDENTIAL Copyright by Spacetek Technology AG. Für dieses Dokument behalten wir uns alle Rechte vor. We reserve all rights in respect of this document.

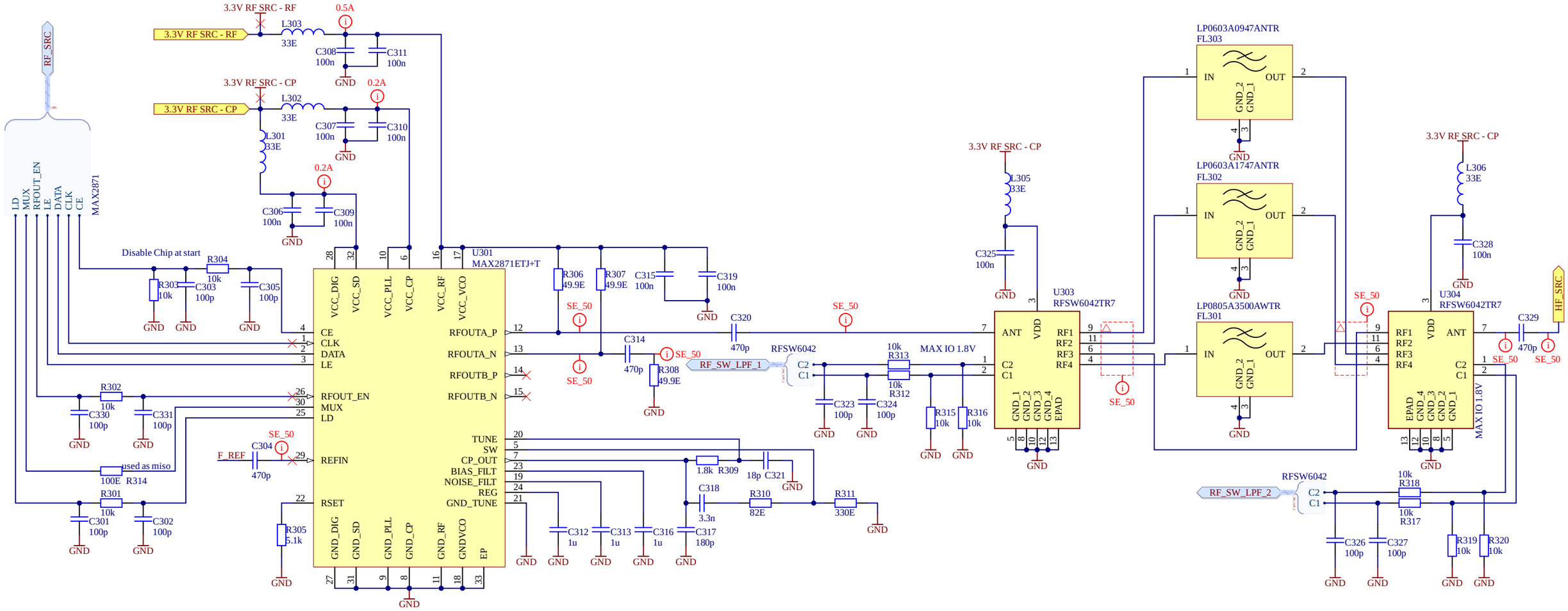
Spacetek Technology AG | Gibb

Spacetek, Gümligen
Gibb, Bern

Project:	SNA	**Number:**	SNA	**Git:**	77bd9c41 [Locally M
Board:	SNA	**Variant:**	EMS	**Version:**	1.0
Page:	MCU	**Release:**	03.09.2026	**Sheet**	3 of 7
				Size:	A3
				Design:	ML
				Check:	RM

File: https://github.com/ml3rc/SNA.git 2 MCU.SchDoc

See LibreVNA Schematics for reference



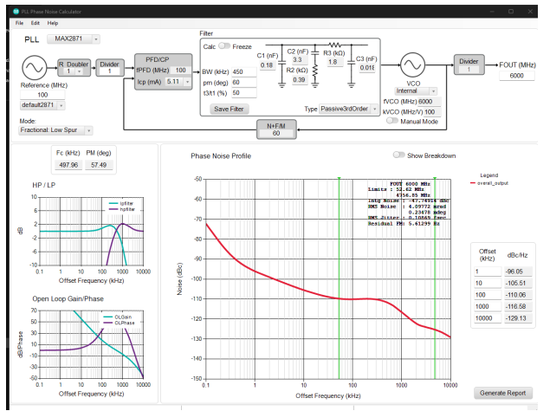
Rset = 5.1kOhm as per datasheet

C_BIAS_FILT, C_NOISE_FILT and C_REG = 1uF as per datasheet

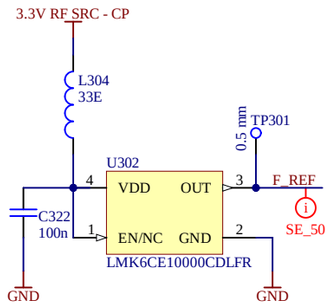
The loop filter can be calculated with:
https://www.analog.com/en/resources/evaluation-hardware-and-software/embedded-development-software/software-download.html?swpart=SFW0007420A

f_bandwidth_loopfilter = ~450kHz -> fast response, but higher noise

R_Crf_coupling = ((1)/(2*π*50MHz*470pF)) = 6.8Ohm

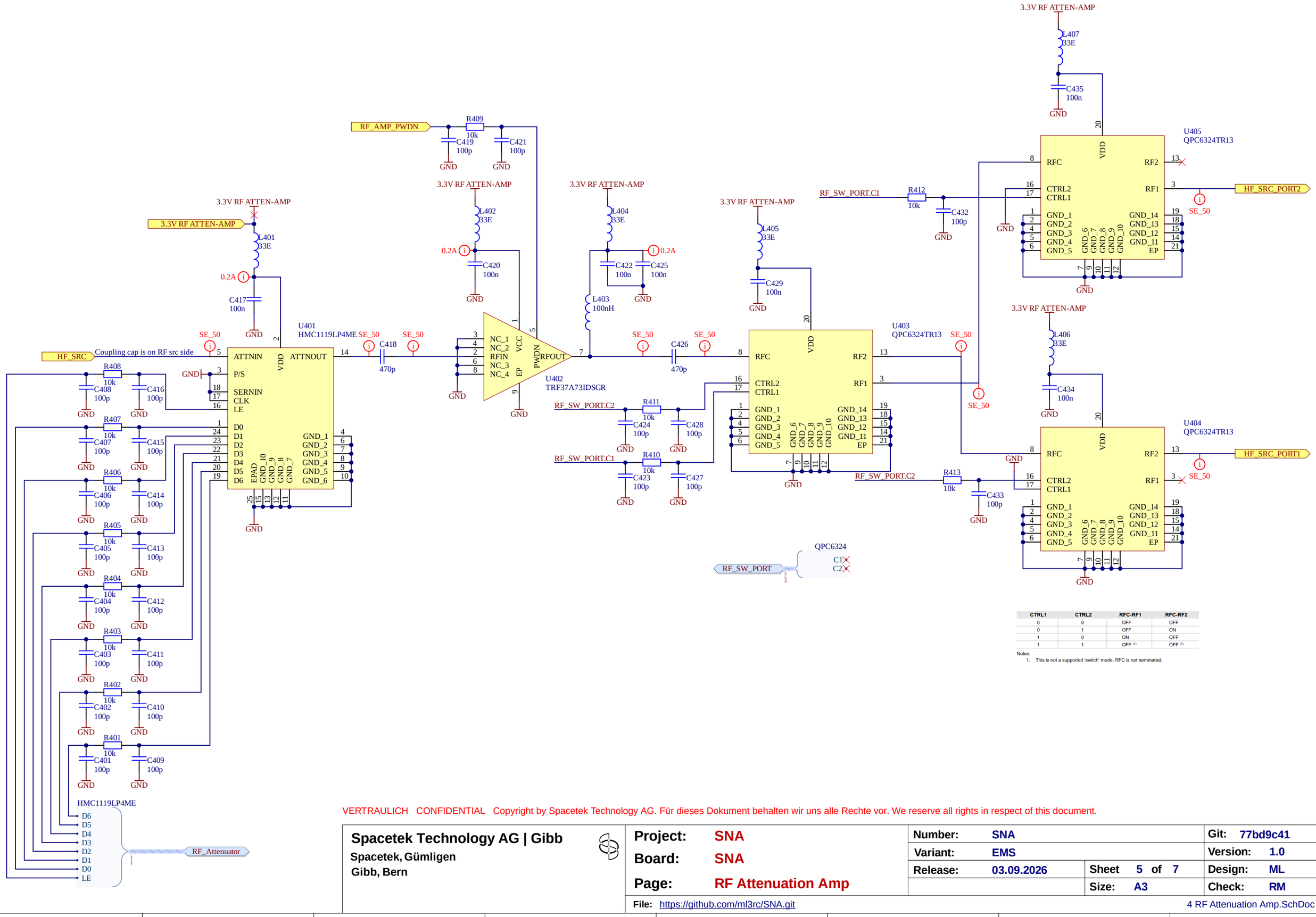


100MHz REF Clk.



VERTRAULICH CONFIDENTIAL Copyright by Spacetek Technology AG. Für dieses Dokument behalten wir uns alle Rechte vor. We reserve all rights in respect of this document.

Spacetek Technology AG | Gibb Spacetek, Gümligen Gibb, Bern	**Project:** SNA **Board:** SNA **Page:** RF Source	**Number:** SNA **Variant:** EMS **Release:** 03.09.2026	**Sheet** 4 of 7 **Size:** A3	**Git:** 77bd9c41 **Version:** 1.0 **Design:** ML **Check:** RM
		File: https://github.com/ml3rc/SNA.git		3 RF Source.SchDoc



CTRL1	CTRL2	RFC-RF1	RFC-RF2
0	0	OFF	OFF
0	1	OFF	ON
1	0	ON	OFF
1	1	OFF (1)	OFF (1)

Notes:
1. This is not a supported 'switch' mode. RFC is not terminated.

VERTRAULICH CONFIDENTIAL Copyright by Spacetek Technology AG. Für dieses Dokument behalten wir uns alle Rechte vor. We reserve all rights in respect of this document.

Spacetek Technology AG | Gibb
Spacetek, Gümligen
Gibb, Bern

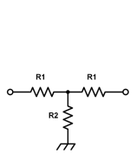
Project:	SNA	**Number:**	SNA	**Git:**	77bd9c41
Board:	SNA	**Variant:**	EMS	**Version:**	1.0
Page:	RF Attenuation Amp	**Release:**	03.09.2026	**Sheet**	5 of 7
				Size:	A3
				Design:	ML
				Check:	RM

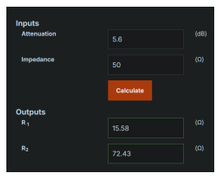
File: https://github.com/ml3rc/SNA.git 4 RF Attenuation Amp.SchDoc

P_r50e_max -> 12dBm -> 15.85mW

BW_output = 5kHz
Cflt = 20nF

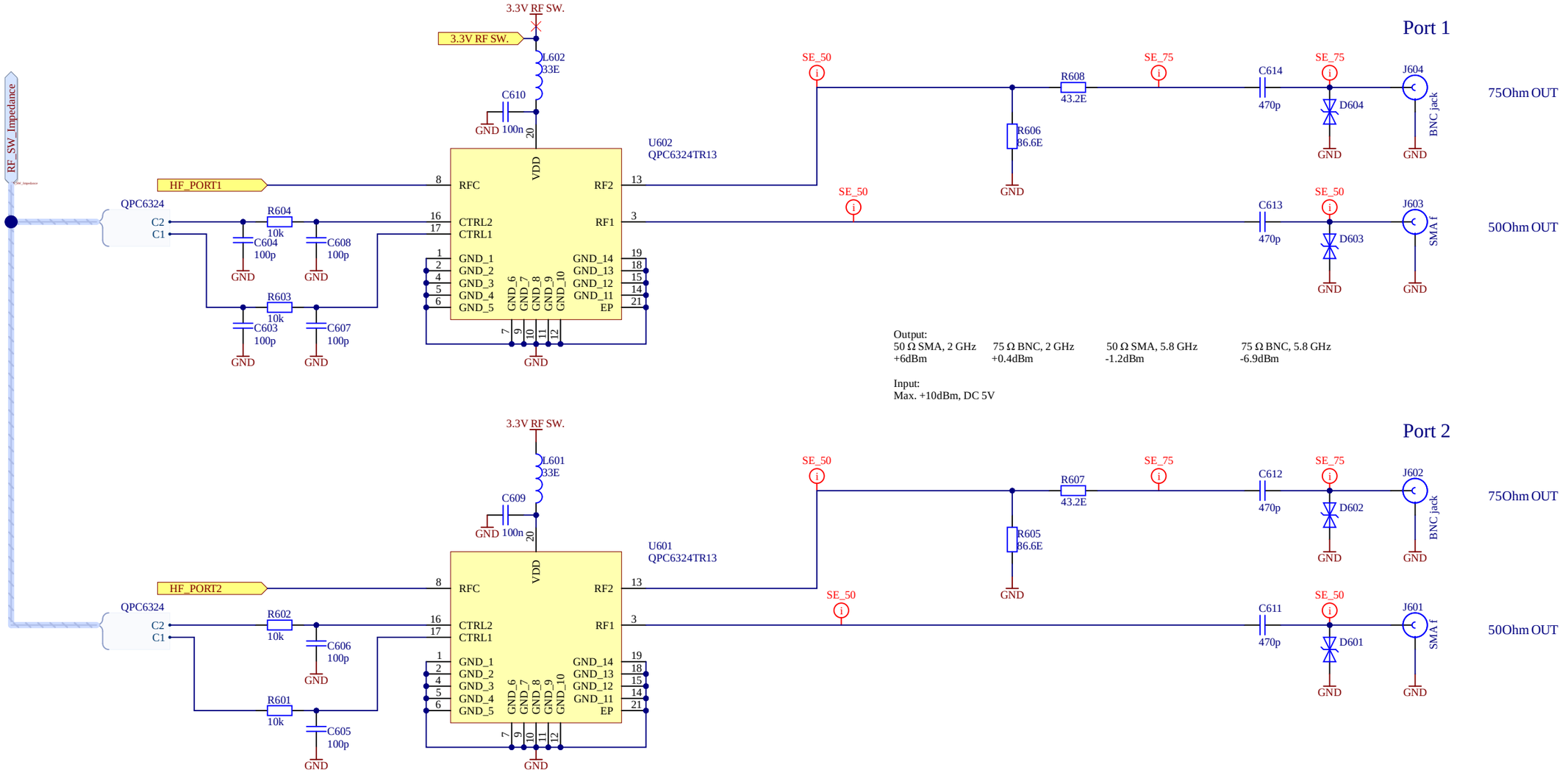
T-Pad calculated with: https://www.allaboutcircuits.com/tools/t-pad-attenuator-calculator/





VERTRAULICH CONFIDENTIAL Copyright by Spacetek Technology AG. Für dieses Dokument behalten wir uns alle Rechte vor. We reserve all rights in respect of this document.

Spacetek Technology AG | Gibb Spacetek, Gümligen Gibb, Bern	**Project:** SNA **Board:** SNA **Page:** RF Directional Coupler	**Number:** SNA **Variant:** EMS **Release:** 03.09.2026	**Git:** 77bd9c41 **Version:** 1.0 **Sheet** 6 **of** 7 **Size:** A3 **Design:** ML **Check:** RM
		File: https://github.com/ml3rc/SNA.git	5 RF Directional Coupler.SchDoc



50-75 Matching Pad:
solve({75=((1)/(((1)/(Rp))+((1)/(50))))+Rs,50=((1)/(((1)/(75+Rs))+((1)/(Rp)))))}
Rp = 86.6Ohm
Rs = 43.3Ohm
Insertion Loss = 5.72dB

VERTRAULICH CONFIDENTIAL Copyright by Spacetek Technology AG. Für dieses Dokument behalten wir uns alle Rechte vor. We reserve all rights in respect of this document.

Spacetek Technology AG | Gibb
Spacetek, Gümligen
Gibb, Bern

Project:	SNA	**Number:**	SNA	**Git:**	77bd9c41
Board:	SNA	**Variant:**	EMS	**Version:**	1.0
Page:	RF Ports	**Release:**	03.09.2026	**Sheet**	7 of 7
				Design:	ML
				Size:	A3
				Check:	RM

File: https://github.com/ml3rc/SNA.git 6 RF Ports.SchDoc